MTH 446: Lecture # 6

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Lecture Span

1. Sets

From last lecture, we claimed that:

Proposition 0.1. *A set F is closed if and only if $\mathbb{C}/F$ is open.*

Proof. (Let F be a closed set) Let $z_0 \in \mathbb{C}/F \iff z_0 \notin F \implies z_0 \notin \partial F$. Next, $z_0 \notin \partial F$ if and only if (z_0 is an interior point of F) or (z_0 is an exterior point of F). Clearly, since $z_0 \in F$, we have that z_0 is an exterior point, and therefore, z_0 is an interior point of $\mathbb{C}/F$. □

Proof. ($\Longleftarrow$) Let $z_0 \in \mathbb{C}/F$, then z_0 is an interior point of $\mathbb{C}/F$. This is the same that z_0 is an exterior point of F. Thus, $z_0 \notin \partial F$. Therefore, $z_0 \notin F$ and $z_0 \notin \partial F$. Then, $\partial F \subseteq F$. □

Definition 0.2. *Let $A \subseteq \mathbb{C}$. Then,*

1. *The interior of A is $A/\partial A$, i.e., the set of all interior points*
2. *The closure of A is $A \cup \partial A$.*

Then, interior of A is int(A) and the closure of A is $\overline{A}$.

Therefore, $\partial A = \overline{A}/\text{int}(A)$. For every set $A \subseteq \mathbb{C}$, then int(A) is open in $\mathbb{C}$. For every set $A \subseteq \mathbb{C}$, then $\overline{A}$ is closed in $\mathbb{C}$.

int(A) is the largest open set containing A, and int(A) is the union of all open sets in A.

Example

ϵ-neighborhood of z_0, $D_{\epsilon_0}(z_0)$. Find $\partial(D_{\epsilon_0}(z_0))$, $\text{int}(D_{\epsilon_0}(z_0))$, and $\overline{D_{\epsilon_0}(z_0)}$.

Solution: We first find ∂D. Then let $z_1 \in D_{\epsilon_0}(z_0)$. Let z_2 be outside of D, then $\exists \epsilon_1 > 0$ such that $D_{\epsilon_1}(z_0) \subseteq \mathbb{C}/D$. Then z_1 is an exterior point of A (outside of D). Then $z_1 \notin \partial A$.

Next, let $z_2 \in D_{\epsilon_0}(z_0)$, then $\exists \epsilon_2$ such that $D_{\epsilon_2}(z_0) \subseteq A = D_{\epsilon_0}(z_0)$. Then, z_2 is an interior point of A, then $z_2 \notin \partial A$.

Let $|z_3 - z_0| = \epsilon_0$. Then, $\forall \epsilon > 0$, $D_\epsilon(z_3) \cap A \neq \emptyset$ and $D_\epsilon(z_3) \cap \mathbb{C}/A \neq 0$. Then, $z_3 \in \partial D$. Then,

$$\partial A = \{|z - z_0| = \epsilon_0\} \tag{1}$$
$$\text{int}(A) = A/\partial A = \{z \in \mathbb{C} : |z - z_0| < \epsilon_0\}/\{z \in \mathbb{C} : |z - z_0| = \epsilon\} \tag{2}$$
$$\overline{A} = A \cup \partial A = \{z \in \mathbb{C} : |z - z_0| \leq \epsilon_0\} \tag{3}$$

Key point: find the boundary point first before finding the other spaces.

Example

Let $U = \{z \in \mathbb{C} : \text{Im}(z) > 1\}$. This is a half plate with a "dotted line" over $y = 1$. Let z_1 be clearly outside of the boundary, then $\exists \epsilon_1 > 0$ such that $D_{\epsilon_1}(z_1) \subseteq \mathbb{C}/U$. Then z_1 is an exterior point of U.

Next, consider z_2 that is clearly in U. Then, exists ϵ_2 such that $D_{\epsilon_2} \subseteq U$. Therefore, z_2 is an interior point of U.

Finally, consider z_2 that lies on the boundary of U, where $y = 1$. Then for all $\epsilon_3 > 0$, then $D_{\epsilon_3}(z_0) \cap U \neq \emptyset$ and $D_{\epsilon_3} \cap (U/\mathbb{C}) \neq \emptyset$. Therefore, $z_3 \in \partial U$. From this, we can define

$$\partial U = \{z \in \mathbb{C} : \text{Im}(z) = 1\} \tag{4}$$

$$\text{int}(U) = U/\partial U = U, \quad \text{thus, } U \text{ is open in } \mathbb{C} \tag{5}$$

$$\overline{U} = U \cup \partial U \tag{6}$$

Example

$G = \{z \in \mathbb{C}/\{0\} : 0 \leq \arg(z) \leq \pi/4\}$. Find ∂G, int(G), and $\overline{G}$.

We can consider three situations. z is clearly inside G, z is clearly outside G, and z is on the edge of G.

Firstly, let z be clearly outside of G. Then $\exists \epsilon_1 > 0$ such that $D_{\epsilon_1}(z_1) \subseteq \mathbb{C}/G \implies z = z_1$ is an exterior point of G. Therefore, $z \notin \partial G$.

Let $z \in G/\{z \in \mathbb{C}/\{0\} : \arg(z) = 0, \pi/4\}$. Then, exists $\epsilon_2 > 0$ such that $D_{\epsilon_2}(z_2) \subseteq G \implies z_2$ is an interior point of $G \implies z_2 \notin \partial G$.

Finally, let $z = 0$. Then for all ϵ, we have that $D(z) \cap G \neq 0$ and $D(z) \cap \mathbb{C}/G \neq \emptyset$. Therefore, $0 \in \partial G$.

On the exam: Use a sketch and also definitions together.

Also, $z_3 \in \{z \in \mathbb{C}/\{0\} : \arg(z) = \pi/4\}$. Then, for all ϵ, $D_\epsilon(z) \cap G \neq 0$ and $D_\epsilon(z) \cap \mathbb{C}/G \neq \emptyset$. Therefore, $z \in \partial G$. Clearly, $z_4 \in \{z \in \mathbb{C}/\{0\} : \arg(z) = 0\} \in \partial G$. Therefore,

$$\partial G = \{0\} \cup \{z \in \mathbb{C}/\{0\} : \arg(z) = \pi/4\} \cup \{z \in \mathbb{C}/\{0\} : \arg(z) = 0\} \tag{7}$$

$$\text{int}(G) = G/\partial G = \text{remove } 0\text{, remove } \pi/4 \text{ and } 0., \quad 0 < \arg(z) < \pi/4 \tag{8}$$

$$\overline{G} = G \cup \partial G = G \cup \{0\} \tag{9}$$

G is not open, since it doesn't contain its boundary points.